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# A simplified and efficient weakly-compressible FV-WENO scheme for immiscible two-phase flows

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ABSTRACT

The present paper proposes a simplified and efficient FV-WENO scheme for simulating weakly-compressible multi-phase flows. Comparing with a classic FV-WENO scheme, the computational accuracy and efficiency are improved by two choices: (i) reconstructing the primitive variables instead of the conservative ones; (ii) performing the WENO reconstruction only once per direction. Although these choices lead to a formally 2nd-order accurate scheme, it is still different from conventional 2nd-order schemes and can be expected to provide higher-order accuracies far from interfaces. An HLLC-type Riemann flux solver is proposed, which appears to be more robust than the linearized Riemann flux solver, allowing the use of smaller artificial sound speeds. The proposed scheme is proven to have the oscillation-free property in the advection of isolated interfaces. A series of validation test-cases have been carried out, in which a good agreement can be observed with references. It is shown that the proposed scheme can give accurate results with little numerical diffusion in the simulations involving weakly-compressible multi-phase flows.

## 1. Introduction

Weakly-compressible (or nearly-incompressible) immiscible multi-phase flows can be often observed in nature and industrial processes, e.g. the oceanic free-surface flows involving air–water interactions and the sloshing in the fuel tank of a vehicle or aircraft, etc. Using computational fluid dynamics (CFD) tools to investigate this type of fluid flows has become nowadays a common way playing an important role as an alternative to experimental studies. Hence, it is of primary importance to develop accurate and efficient numerical schemes for simulating this type of multi-phase flows.

In weakly-compressible fluid flows, the compressibility in fluids is usually not of interest, and thus can be completely ignored or removed. The existing CFD methods for simulating this type of fluid flows can be classified into two categories: incompressible scheme and weakly-compressible scheme. In an incompressible scheme, the mass conservation equation based on the density is replaced by the zero-divergence condition for the velocity field. The correction part of the new velocity of the entire fluid domain is updated simultaneously after solving the Poisson-pressure equation. In most cases, the incompressible scheme is adequate and has been widely applied for simulating multi-phase flows with the help of an interface capturing or tracking technique, such as the level-set (LS) [1,2] or the volume-of-fluid (VOF) [3] methods, etc. A major advantage of the incompressible scheme is that the implicit feature (when obtaining the correction of velocity) allows one to use a larger time-step. However, solving the Poisson-pressure equation needs to handle a large linear system, which requires special parallelization algorithms and might encounter sometimes difficulties in the iterative convergence procedure.

As for the weakly-compressible scheme, there is no need to solve the Poisson-pressure equation and the time integration is often carried out in an explicit way. This feature simplifies largely the structure of the computational code and allows a better scalability for parallelization with many CPUs [4]. In addition, it is easier to extend the code for simulating compressible flows by adding the equation of energy, while maintaining the resolution procedure. However, one has to choose *a priori* an appropriate artificial sound speed in the fluid, which is supposed to be large enough (comparing with the maximal fluid velocity) for verifying the weakly-compressible hypothesis, and is usually expected to be small enough for having a reasonable time-step due to the CFL condition. In fact, this idea of using an artificial sound speed is adopted in various types of CFD methods, such as the smoothed particle hydrodynamics (SPH) method [5–8], lattice Boltzmann method [9], etc. It is worth noting here that it is difficult to assert which type of scheme (incompressible or weakly-compressible) is better, as the conclusion depends on the intended applications. For more analysis and discussions, one may refer to the comparison work in [10].

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In the literature, several weakly-compressible multi-phase schemes [11–15] have been proposed for various types of applications. In our previous work [16], we proposed a weakly-compressible finite volume (FV) scheme based on the weighted essentially non-oscillating (WENO) [17] reconstruction method for simulating immiscible multi-phase flows. A three-equation mathematical model with instantaneous pressure equilibrium was adopted, which is sufficient for describing weakly-compressible two-phase flows. The conservative variables were reconstructed with the WENO scheme at each cell-interface for computing the flux with a linearized Riemann flux solver. Although this scheme is 5th-order accurate in the convergence study, it might suffer from severe numerical oscillations in some violent cases, especially when there is only one available stencil (e.g. near the solid wall). In addition, there are more than one 'sweep' in the WENO reconstruction, which is computationally quite costly.

Based on the above observations, in the present work, we choose to reconstruct the primitive variables instead of the conservative ones, because in the aimed applications, such as air–water interaction flows, only the volume fraction field possesses steep gradients in most cases. With this choice, more stencils are activated when reconstructing the other smoother primitive variables (e.g. the phase-density and the velocity components) near the phase-interface, which may reduce the numerical diffusion. However, unlike the reconstruction scheme in [18], we propose to get the cell-average primitive variables in a straightforward way for the sake of computational efficiency. Although this makes the numerical scheme formally 2nd-order accurate in space, as shown in Appendix A, a higher-order accuracy can still be expected at the places far from the phase-interface.

Besides, in the present work, we choose to perform the WENO reconstruction with only one 'sweep' per direction, which can be considered as there is only one Gauss quadrature point at each cell-interface. In general, this choice gives a formally 2nd-order accuracy. Nevertheless, when the governing equations become almost linear, the flux term can still be computed with a higher-order accuracy, even with only one 'sweep' per direction in the WENO reconstruction.

As a consequence, thanks to the strategies presented above, the computational efficiency can be largely improved in the proposed multi-phase FV-WENO scheme. It is noteworthy that, although the proposed scheme is formally 2nd-order accurate in space, it is different from a conventional 2nd-order scheme such as the MUSCL one [19], because a higher-order accuracy can still be obtained at the places far from the phase-interface, where the mixture's density is almost constant or the governing equation is almost linear. Furthermore, an HLLC-type Riemann flux solver is proposed in the present work for the adopted hyperbolic model and it can be proven that the proposed scheme induces strictly zero oscillation in the advection of an isolated interface test-case [20]. It is observed that the HLLC-type Riemann flux solver appears to be more robust than the linearized one in the sense that smaller artificial sound speeds can be used.

The rest of the paper is organized as follows. Section 2 presents the adopted mathematical model and the FV formulation. The details about the WENO reconstruction scheme such as the reconstructed variables and the number of Gauss points are given in Section 3. The proposed HLLC-type Riemann flux solver is presented in Section 4. In Section 5, it is shown that it is trivial for the proposed scheme to have the zero-oscillation feature. Some validation test-cases have been carried out and shown in Section 6. Finally, the conclusions are drawn in Section 7.

## 2. Mathematical model and finite volume formulation

### 2.1. Equilibrium model for multi-phase weakly-compressible flows

As done in our previous work [16], to simulate weakly-compressible two-phase flows, we choose the mathematical model based on three-equations [21,22], which is given as

$$\begin{cases} \dfrac{\partial(\alpha_1\rho_1)}{\partial t} + \nabla\cdot\left(\alpha_1\rho_1\mathbf{u}\right) = 0, \\ \dfrac{\partial(\alpha_2\rho_2)}{\partial t} + \nabla\cdot\left(\alpha_2\rho_2\mathbf{u}\right) = 0, \\ \dfrac{\partial(\rho\mathbf{u})}{\partial t} + \nabla\cdot(\rho\mathbf{u}\otimes\mathbf{u} + p\mathbf{I}) = \rho\mathbf{g}, \end{cases} \tag{1}$$

where $\alpha_k$ and $\rho_k$ denote respectively the volume fraction and density for the $k$th phase with $k = 1$ and 2. By definition, one has $\alpha_1+\alpha_2 = 1$ and the density of the mixture is $\rho = \alpha_1\rho_1 + \alpha_2\rho_2$. Moreover, the pressure of the mixture is defined as $p = \alpha_1 p_1 + \alpha_2 p_2$, where $p_k$ denotes the pressure of the $k$th phase. A barotropic equation of state (EOS) is chosen to relate $p_k$ with $\rho_k$ as $p_k = p_0 + c_k^2(\rho_k - \rho_{k,0})$, where $p_0$, $c_k$ and $\rho_{k,0}$ are three constants denoting the reference pressure, the speed of sound and the reference density for the $k$th phase, respectively. Finally, $\mathbf{u}$ denotes the common velocity field shared by the two phases and $\mathbf{g}$ is the gravity vector.

The above system of Eqs. (1) is called the 'equilibrium model', as the pressure equilibrium between the two phases is instantaneously assured everywhere, i.e. $p_1(\mathbf{x},t) = p_2(\mathbf{x},t)$ with $\mathbf{x}\in\mathbb{R}^d$ and $t\in\mathbb{R}^+$ in a $d$-dimensional problem. As a consequence, the sound speed $c$ of the two-phase mixture is given as

$$\frac{1}{\rho c^2} = \frac{\alpha_1}{\rho_1 c_1^2} + \frac{\alpha_2}{\rho_2 c_2^2}. \tag{2}$$

### 2.2. Finite volume formulation

In order to solve numerically the governing Eqs. (1), we adopt the finite volume (FV) method for the spacial discretization. To do so, Eq. (1) can be rewritten in an integral form over the fixed elemental control volume $\Omega_{i,j}$ as

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_{i,j}} \mathbf{U}\,\mathrm{d}\Omega + \int_{\partial\Omega_{i,j}} \mathbf{F}\,\mathrm{d}\Gamma = \int_{\Omega_{i,j}} \mathbf{G}\,\mathrm{d}\Omega, \tag{3}$$

with

$$\mathbf{U} = \begin{bmatrix} \alpha_1\rho_1 \\ \alpha_2\rho_2 \\ \rho u \\ \rho v \end{bmatrix}, \mathbf{F} = \begin{bmatrix} \alpha_1\rho_1(\mathbf{n}\cdot\mathbf{u}) \\ \alpha_2\rho_2(\mathbf{n}\cdot\mathbf{u}) \\ \rho u(\mathbf{n}\cdot\mathbf{u}) + n_x p \\ \rho v(\mathbf{n}\cdot\mathbf{u}) + n_y p \end{bmatrix} \text{ and } \mathbf{G} = \begin{bmatrix} 0 \\ 0 \\ \rho g_x \\ \rho g_y \end{bmatrix}, \tag{4}$$

where $u$, $v$, $n_x$, $n_y$, $g_x$ and $g_y$ denote the components of the velocity vector $\mathbf{u}$, the outward-pointing normal vector $\mathbf{n}$ at the boundary $\partial\Omega_{i,j}$ and the gravity vector in $x$- and $y$-directions, respectively. Notice that although the above formulation is expressed in two-dimensions, it is straightforward to extend to three-dimensional cases.

After the spacial discretization with a fixed Cartesian mesh, the semi-discrete system can be written as

$$\frac{\mathrm{d}\overline{\mathbf{U}}_{i,j}}{\mathrm{d}t} = -\frac{1}{\Delta V_{i,j}}\sum_e \widehat{\mathbf{F}}_e \Delta S_e + \overline{\mathbf{G}}_{i,j}, \tag{5}$$

where $\overline{\mathbf{U}}_{i,j}$ and $\overline{\mathbf{G}}_{i,j}$ denote the cell-average values of $\mathbf{U}$ and $\mathbf{G}$, respectively, defined as

$$\overline{\mathbf{U}}_{i,j} = \frac{1}{\Delta V_{i,j}}\int_{\Omega_{i,j}} \mathbf{U}\,\mathrm{d}\Omega = \frac{1}{\Delta x\Delta y}\int_{x_{i-1/2}}^{x_{i+1/2}}\int_{y_{j-1/2}}^{y_{j+1/2}} \mathbf{U}\,\mathrm{d}y\,\mathrm{d}x, \tag{6}$$

with $\Delta V_{i,j} = \Delta x\Delta y$ being the volume of the FV-cell.

Besides, in Eq. (5), $\widehat{\mathbf{F}}_e$ denotes the face-average value of the flux $\mathbf{F}$ over the $e$th boundary of the control volume $\Omega_{i,j}$, which has a surface

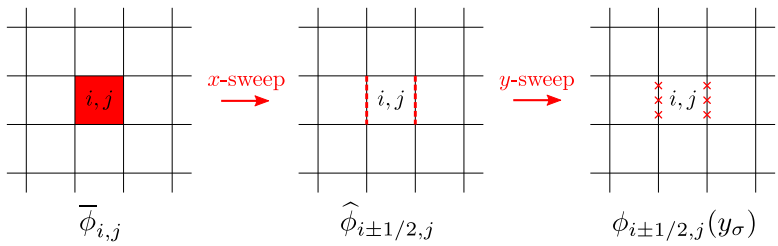


**Fig. 1.** Conventional WENO reconstruction procedure in a 2D example.

area of $\Delta S_e$. The value of $\widehat{\mathbf{F}}_e$ can be approximately obtained by means of a Gauss quadrature over the cell-boundary. In order to carry out the Gauss quadrature, we need to reconstruct the variables at cell-interfaces first, and then get the flux by using an approximate Riemann flux solver.

## 3. WENO reconstruction

In the present work, we apply the 5th-order WENO scheme [17] to reconstruct the values of variables at cell interfaces. The major steps of the reconstruction are briefly presented subsequently and more details can be found in some previous works such as in [18,23].

For example, in a 2D case as shown in Fig. 1, the WENO scheme usually proceeds in two steps: (1) with the cell-average value $\overline{\phi}_{i,j}$, the WENO reconstruction in $x$-direction ($x$-sweep) gives the face-average value $\widehat{\phi}_{i\pm1/2,j}$ at $x_{i\pm1/2}$; (2) after doing this for all cells, the WENO reconstruction in $y$-direction ($y$-sweep) gives then the value $\phi_{i\pm1/2,j}(y_\sigma)$ at each Gauss quadrature point located at $x_{i\pm1/2}$ and $y_\sigma$.

In order to obtain the desired convergence order, one must pay attention to the choice of the reconstructed variables and the number of Gauss points. Next, we shall discuss the influences of these two aspects and give the reasons for the strategies that we propose in the present paper.

### 3.1. Reconstructed variables

It should be noted that only the cell-average conservative variables $\overline{\mathbf{U}}_{i,j}$ are available at the beginning of each time-step. Hence, it is straightforward to reconstruct the conservative variables $\mathbf{U} = [\alpha_1\rho_1, \alpha_2\rho_2, \rho u, \rho v]^\top$. However, although this choice can maintain the convergence order, it is often not easy to avoid spurious pressure oscillations near phase-interfaces, even in the case with uniform pressure and velocity fields [16,18].

Usually, reconstructing the primitive variables may lead to less oscillation [24]. However, if one directly reconstructs the primitive variables, the convergence order in space is limited to 2nd-order in 2D cases, even though it can attain the desired higher order convergence in 1D cases [18].

In our previous work [16], a common smoothness indicator has been adopted in the WENO reconstruction procedure, which allowed one to maintain the high convergence order, while avoiding pressure oscillation in the advection of isolated phase-interfaces. Despite these improvements, the WENO scheme in [16] still reconstructed the conservative variables.

In the context of simulating immiscible two-phase flows with a diffuse interface model, it is usually expected to maintain the phase-interface as sharp as possible. In other words, the volume fraction field $\alpha_1$ usually varies significantly across the phase-interface. As a consequence, all the conservative variables possess steep gradients, even though $p$ ($\rho_1$ or $\rho_2$), $u$ and $v$ might be smooth near the phase-interface.

When using the WENO scheme to reconstruct the conservative variables near the phase-interface, it is often seen that only one stencil is activated for all reconstructed variables, just due to the steep gradient in the $\alpha_1$ field, which may degrade the numerical accuracy of the scheme.

Based on the previous observations and analysis, in the present work, we propose to reconstruct the primitive variables $\mathbf{W} = [\alpha_1, \rho_1, u, v]^\top$, which possesses the following advantages:

- Among all the reconstructed variables, only $\alpha_1$ has a steep gradient across the phase-interface, especially in weakly-compressible flows;
- With the equilibrium model, it suffices to reconstruct $\rho_1$, because $\rho_2$ and $p$ can be immediately computed using the adopted EOS, once $\rho_1$ is obtained;
- It is trivial to strictly avoid spurious oscillations of pressure in the advection of isolated interface [24].

However, it must be noted that there is a major drawback, which is related to the fact that the cell-average values of the primitive variables are not directly available. In [18], it was proposed to compute the cell-average primitive variables from the cell-average conservative variables by means of the WENO reconstruction. In the present work, we choose to obtain the cell-average primitive variables in the following way for the sake of computational efficiency.

#### 3.1.1. Volume fraction field $\alpha_1$

In the adopted equilibrium model [16,22], the value of $\alpha_1$ is obtained from the instantaneous pressure equilibrium condition $p_1 = p_2$. After each time-integration, once $\overline{(\alpha_1\rho_1)}_{i,j}$ and $\overline{(\alpha_2\rho_2)}_{i,j}$ are updated, we propose to get the value of $\overline{(\alpha_1)}_{i,j}$ by solving the pressure equilibrium equation

$$c_1^2\left(\frac{\overline{(\alpha_1\rho_1)}_{i,j}}{\overline{(\alpha_1)}_{i,j}} - \rho_{1,0}\right) = c_2^2\left(\frac{\overline{(\alpha_2\rho_2)}_{i,j}}{1-\overline{(\alpha_1)}_{i,j}} - \rho_{2,0}\right), \tag{7}$$

which can be proven to have only one valid solution.

#### 3.1.2. Density field $\rho_1$

Once the cell-average volume fraction $\overline{(\alpha_1)}_{i,j}$ is obtained, we propose to compute the value of the cell-average density $\overline{(\rho_1)}_{i,j}$ as

$$\overline{(\rho_1)}_{i,j} \simeq \widetilde{(\rho_1)}_{i,j} = \frac{\overline{(\alpha_1\rho_1)}_{i,j}}{\overline{(\alpha_1)}_{i,j}}, \tag{8}$$

where $\widetilde{(\rho_1)}_{i,j}$ is an approximation of the cell-average value $\overline{(\rho_1)}_{i,j}$.

#### 3.1.3. Velocity components $u$ and $v$

Similarly, the cell-average values of the velocity components can be obtained as

$$\overline{(u)}_{i,j} \simeq \widetilde{(u)}_{i,j} = \frac{\overline{(\rho u)}_{i,j}}{\overline{(\rho)}_{i,j}} \quad \text{and} \quad \overline{(v)}_{i,j} \simeq \widetilde{(v)}_{i,j} = \frac{\overline{(\rho v)}_{i,j}}{\overline{(\rho)}_{i,j}}, \tag{9}$$

where the cell-average value of the mixture's density can be directly computed as $\overline{(\rho)}_{i,j} = \overline{(\alpha_1\rho_1)}_{i,j} + \overline{(\alpha_2\rho_2)}_{i,j}$.

It is here worth noting that, although the operations in Eqs. (7)–(9) are simple and straightforward, they generally introduce an error of order $\mathcal{O}(\Delta x^2)$ as shown in Appendix A. Nevertheless, when $\alpha_1$ and $\rho_1$ are almost constant, e.g. at the places far from the phase-interface, a higher-order of accuracy can still be maintained.

### 3.2. Number of Gauss points

As shown in Eq. (5), a FV scheme requires to evaluate numerically the face-average flux $\widehat{\mathbf{F}}_e$. Taking the Cartesian mesh in Fig. 1 as an example, the face-average flux at $x_{i+1/2}$ is defined and calculated as

$$\widehat{\mathbf{F}}_{i+1/2,j} = \frac{1}{\Delta y}\int_{y_{j-1/2}}^{y_{j+1/2}} \mathbf{F}(x_{i+1/2}, y)\,\mathrm{d}y \simeq \frac{1}{\Delta y}\sum_{\sigma=1}^{N_\sigma} w_\sigma \mathbf{F}_{i+1/2,\sigma}, \tag{10}$$

where $N_\sigma$ denotes the total number of Gauss quadrature points, $w_\sigma$ the quadrature weighting coefficient and $\mathbf{F}_{i+1/2,\sigma} = \mathbf{F}(x_{i+1/2}, y_\sigma)$.

It has been pointed out in [25] that at least three Gauss quadrature points are needed to maintain the 5th-order accuracy of the WENO scheme. In some other works [16,18,23], it was observed that two Gauss points are sufficient to obtain the 5th-order accuracy. These require more than one sweep in the WENO reconstruction for each direction, as shown in Fig. 1.

In the present work, we propose to carry out the WENO reconstruction of the primitive variables $\mathbf{W}$ only once per direction, i.e. with only one sweep per direction. For instance, as shown in Fig. 1, after the $x$-sweep, we can obtain the face-average value $\widehat{\mathbf{W}}_{i+1/2,j}$ at the boundary located at $x_{i+1/2}$ over the interval $y \in [y_{j-1/2}, y_{j+1/2}]$. We propose to directly use this $\widehat{\mathbf{W}}_{i+1/2,j}$ in the approximate Riemann flux solver, which will be presented in the next section, that is to use only one Gauss point.

Although this choice renders the scheme formally 2nd-order accurate, it is computationally less demanding, as it avoids the WENO reconstruction in $y$-direction ($y$-sweep). Moreover, when the Jacobian matrix $\mathbf{A} = \partial\mathbf{F}/\partial\mathbf{W}$ is almost constant, the face-average flux $\widehat{\mathbf{F}}_{i+1/2}$ can then be expressed as

$$\widehat{\mathbf{F}}_{i+1/2,j} = \frac{1}{\Delta y}\int_{y_{j-1/2}}^{y_{j+1/2}} \mathbf{F}(x_{i+1/2}, y)\,\mathrm{d}y = \frac{\mathbf{A}}{\Delta y}\int_{y_{j-1/2}}^{y_{j+1/2}} \mathbf{W}(x_{i+1/2}, y)\,\mathrm{d}y = \mathbf{A}\widehat{\mathbf{W}}_{i+1/2,j}, \tag{11}$$

which means that the proposed scheme is still of higher-order accuracy, when $\mathbf{A}$ is heuristically observed to be almost constant and $\widehat{\mathbf{W}}_{i+1/2,j}$ is reconstructed with higher-order accuracy.

In summary, in the present work, we propose to reconstruct the primitive variables only once per direction with the WENO scheme, which leads to less computational overhead. Although it is formally 2nd-order accurate, in some special cases, e.g. at the places far from the phase-interface, the proposed scheme can still be of higher-order accuracy. Hence, it can be expected to provide more accurate results than some conventional 2nd-order accurate schemes such as the MUSCL one [19], etc.

## 4. HLLC-type Riemann flux solver

Once the primitive variables are reconstructed on the left-hand and right-hand sides of the cell-interface, we can compute the flux using an approximate Riemann flux solver. To do so, in the present work, based on the adopted equilibrium model (1), we propose an HLLC-type Riemann flux solver [26]. To estimate the numerical flux at the boundary orthogonal to $x$-direction (e.g. at $x_{i+1/2}$), one usually analyzes the quasi-linear $x$-split governing equations. As shown in [16], the hyperbolic part of the governing equations possesses four eigenvalues $\lambda^{(1)} = u - c$, $\lambda^{(2)} = \lambda^{(3)} = u$ and $\lambda^{(4)} = u + c$, corresponding to three waves of speed $S_L$, $S_M$ and $S_R$ in the $x - t$ plane, as shown in Fig. 2.

The Rankine–Hugoniot condition can be written across the $S_L$-wave or $S_R$-wave as

$$\mathbf{F}_X^* = \mathbf{F}_X + S_X(\mathbf{U}_X^* - \mathbf{U}_X), \tag{12}$$

or explicitly as

$$\begin{bmatrix} \alpha_{1,X}^*\rho_{1,X}^*u^* \\ \alpha_{2,X}^*\rho_{2,X}^*u^* \\ \rho_X^*u^*u^* + p^* \\ \rho_X^*v_Xu^* \end{bmatrix} = \begin{bmatrix} \alpha_{1,X}\rho_{1,X}u_X \\ \alpha_{2,X}\rho_{2,X}u_X \\ \rho_Xu_Xu_X + p_X \\ \rho_Xv_Xu_X \end{bmatrix} + S_X\left(\begin{bmatrix} \alpha_{1,X}^*\rho_{1,X}^* \\ \alpha_{2,X}^*\rho_{2,X}^* \\ \rho_X^*u^* \\ \rho_X^*v_X \end{bmatrix} - \begin{bmatrix} \alpha_{1,X}\rho_{1,X} \\ \alpha_{2,X}\rho_{2,X} \\ \rho_Xu_X \\ \rho_Xv_X \end{bmatrix}\right), \tag{13}$$

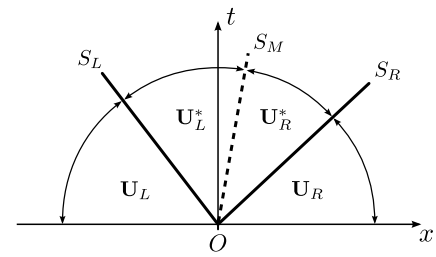


**Fig. 2.** Eigenstructure of the local Riemann problem at each cell-interface with the subscripts '$L$' and '$R$' denoting the left- and right-hand sides of the interface, respectively.

where $X = L$ or $R$, corresponding respectively to the $S_L$-wave or $S_R$-wave.

Once the wave-speeds are estimated as $S_L = \min(u_L - c_L, u_R - c_R)$ and $S_R = \max(u_L + c_L, u_R + c_R)$ [27], we can derive the contact wave-speed $S_M = u^*$ from Eq. (13) as

$$S_M = u^* = \frac{p_R - p_L + \rho_R u_R(u_R - S_R) - \rho_L u_L(u_L - S_L)}{\rho_R(u_R - S_R) - \rho_L(u_L - S_L)}. \tag{14}$$

Additionally, by using the first two equations in Eq. (13), we can obtain

$$\alpha_{k,X}^*\rho_{k,X}^* = \alpha_{k,X}\rho_{k,X}\frac{S_X - u_X}{S_X - S_M}, \qquad \text{with } k = 1,2 \text{ and } X = L, R, \tag{15}$$

which gives then the mixture's density in the star region as $\rho_X^* = \alpha_{1,X}^*\rho_{1,X}^* + \alpha_{2,X}^*\rho_{2,X}^*$. As shown in [28], it can be proved that $\alpha_{k,X}^*\rho_{k,X}^*$ can be ensured to be positive.

Now, the fluid state $\mathbf{U}_X^*$ in the star region is obtained, which allows us to calculate the flux vector $\mathbf{F}_X^*$ directly using Eq. (12). Finally, the HLLC-type Riemann flux vector at $x_{i+1/2}$ is computed as

$$\mathbf{F}_{i+1/2,j}^{\mathrm{HLLC}} = \begin{cases} \mathbf{F}_L, & \text{if } 0 < S_L < S_M < S_R, \\ \mathbf{F}_L^*, & \text{if } S_L < 0 < S_M < S_R, \\ \mathbf{F}_R^*, & \text{if } S_L < S_M < 0 < S_R, \\ \mathbf{F}_R, & \text{if } S_L < S_M < S_R < 0, \end{cases} \tag{16}$$

where the first and fourth cases may be encountered near the diffuse interface, due to small sound speeds in the mixture [16].

Probably due to the feature satisfying the Rankine–Hugoniot condition (jump condition across waves) it was observed numerically that the HLLC-type Riemann flux solver provided more robust simulations than with the linearized Riemann flux solver in [16]. Furthermore, as an additional benefit, it turned out that a smaller artificial sound speed could be used.

## 5. Oscillation-free property

As stated in [20,29], one major difficulty for a numerical multiphase scheme is to avoid non-physical spurious oscillations across the phase-interface. Here, we provide an analysis showing that the proposed scheme can strictly avoid spurious oscillations near phase-interfaces in the advection of isolated interface [20].

To this end, we consider a 1D advection problem with initially uniform pressure and velocity fields, i.e. $p(x,0) = p_0$ and $u(x,0) = u_0$ (e.g. $u_0 > 0$). Meanwhile, the volume fraction field $\alpha_1(x,0)$ can be arbitrarily distributed in space, either smooth or sharp.

With the above initial conditions, the WENO reconstruction of the primitive variables at $x_{i+1/2}$ gives $u_L = u_R = u_0$, $p_L = p_R = p_0$, $\rho_{k,L} = \rho_{k,R} = \rho_{k,0}$ with $k = 1,2$. Consequently, according to Eq. (14), the contact wave-speed $S_M$ is then computed as $S_M = u^* = u_0$. Thus, using Eq. (15), one can obtain $\alpha_{k,X}^*\rho_{k,X}^* = \alpha_{k,X}\rho_{k,0}$.

**Table 1**
Comparison between the present FV-WENO scheme and the previous one proposed in [16].

| | Reconstructed variables | Number of Gauss points | Riemann flux solver |
|---|---|---|---|
| Previous scheme [16] | Conservative variables | Two | Linearized |
| Present scheme | Primitive variables | One | HLLC-Type |

### 5.1. Pressure field

Based on Eq. (16), the HLLC-type Riemann flux at $x_{i+1/2}$ is then $\mathbf{F}^{\mathrm{HLLC}}_{i+1/2} = \mathbf{F}^{*}_{L}$ in the weakly-compressible limits. Similarly, one can also get the flux $\mathbf{F}^{\mathrm{HLLC}}_{i-1/2}$ at $x_{i-1/2}$. As a result, after one explicit time-integration of Eq. (5), the cell-average fluid state can be updated for the $i$th cell as

$$\overline{(\alpha_k\rho_k)}_i^{n+1} = \overline{(\alpha_k\rho_k)}_i^{n} - \frac{\Delta t}{\Delta x}\left(\alpha^{\mathrm{HLLC}}_{k,i+1/2} - \alpha^{\mathrm{HLLC}}_{k,i-1/2}\right)\rho_{k,0}u_0, \tag{17}$$

where $\overline{(\alpha_k\rho_k)}_i^{n} = \overline{(\alpha_k)}_i^{n}\rho_{k,0}$ because of the initially uniform pressure field. In addition, $\alpha^{\mathrm{HLLC}}_{k,i+1/2}$ and $\alpha^{\mathrm{HLLC}}_{k,i-1/2}$ are the volume fraction values in the HLLC-type Riemann flux solver at the cell-interfaces $x_{i+1/2}$ and $x_{i-1/2}$, respectively.

As presented in Section 3.1.1, the volume fraction field $\overline{(\alpha_1)}_i^{n+1}$ is obtained by solving Eq. (7) with the newly updated $\overline{(\alpha_k\rho_k)}_i^{n+1}$ in Eq. (17). Since Eq. (7) possesses a unique solution, as shown in [16], if one can find out a solution $\overline{(\alpha_1)}_i^{n+1}$, it must be the unique solution of Eq. (7).

With the help of Eq. (17), it is easy to verify that $\overline{(\alpha_1)}_i^{n+1} = \overline{(\alpha_1\rho_1)}_i^{n+1}/\rho_{1,0}$ is just the sought solution that assures the instantaneous pressure equilibrium. Finally, as shown in Section 3.1.2, the cell-average density field can then be computed by Eq. (8) as $\overline{(\rho_1)}_i^{n+1} = \rho_{1,0}$, which implies that the pressure field remains constant, or in other words, there is strictly no pressure oscillation.

### 5.2. Velocity field

By summing up the two equations ($k = 1$ and 2) in Eq. (17), one can obtain the updated cell-average density of the mixture as

$$\begin{aligned}\overline{(\rho)}_i^{n+1} &= \overline{(\alpha_1\rho_1)}_i^{n+1} + \overline{(\alpha_2\rho_2)}_i^{n+1},\\ &= \overline{(\alpha_1\rho_1)}_i^{n} + \overline{(\alpha_2\rho_2)}_i^{n} - \frac{\Delta t}{\Delta x}\left(\left(\alpha^{\mathrm{HLLC}}_{1,i+1/2}\rho_{1,0} + \alpha^{\mathrm{HLLC}}_{2,i+1/2}\rho_{2,0}\right)\right.\\ &\quad\left. - \left(\alpha^{\mathrm{HLLC}}_{1,i-1/2}\rho_{1,0} + \alpha^{\mathrm{HLLC}}_{2,i-1/2}\rho_{2,0}\right)\right)u_0,\\ &= \overline{(\rho)}_i^{n} - \frac{\Delta t}{\Delta x}\left(\rho^{\mathrm{HLLC}}_{i+1/2} - \rho^{\mathrm{HLLC}}_{i-1/2}\right)u_0.\end{aligned} \tag{18}$$

Next, as done in Section 5.1, after one explicit time integration with the HLLC-type flux given in Eq. (13), we can have

$$\overline{(\rho u)}_i^{n+1} = \overline{(\rho u)}_i^{n} - \frac{\Delta t}{\Delta x}\left(\rho^{\mathrm{HLLC}}_{i+1/2} - \rho^{\mathrm{HLLC}}_{i-1/2}\right)u_0^2 = \left(\overline{(\rho)}_i^{n} - \frac{\Delta t}{\Delta x}\left(\rho^{\mathrm{HLLC}}_{i+1/2} - \rho^{\mathrm{HLLC}}_{i-1/2}\right)u_0\right)u_0. \tag{19}$$

Consequently, the cell-average velocity can then be computed by Eq. (9) as $\overline{(u)}_i^{n+1} = u_0$, which means that the velocity field remains also constant at next instant with strictly no velocity oscillation.

Hence, we have proven that the proposed numerical scheme can strictly avoid spurious oscillations in the advection of an isolated interface with uniform pressure and velocity fields. Notice that it is trivial for the scheme to have this property, as no extra operation is needed.

Before showing the numerical results, we provide in Table 1 a summary comparing some key features of the present FV-WENO scheme with the one proposed in our previous work [16].

## 6. Validation test-cases

In all the following validation test-cases, the semi-discrete system of Eqs. (5) is integrated in time with the classic 4th-order Runge–Kutta scheme. Due to the explicit feature, the time-step is computed with a CFL number of 0.7 in all tests. Additionally, in all the test-cases, the reference densities of water and air are set as $\rho_{1,0} = 1000\ \mathrm{kg/m^3}$ and $\rho_{2,0} = 1\ \mathrm{kg/m^3}$, respectively. The initial value of the volume fraction $\alpha_1$ is equal to $1 - \varepsilon_\alpha$ in water and $\varepsilon_\alpha$ in air with $\varepsilon_\alpha = 10^{-4}$. Notice that this value of $\varepsilon_\alpha$ should be small enough for ensuring approximate immiscibility, however it cannot be equal to zero for being able to solve Eq. (7).

As the proposed scheme reconstructs the primitive variables only once per direction, all the results obtained with the present scheme are noted as 'Prim-1G', where 'Prim' and '1G' mean the primitive variables and one Gauss quadrature point, respectively. In the meantime, the results are compared with the ones obtained using the scheme based on the reconstruction of the conservative variables at two Gauss points [16], which are noted as 'Cons-2G'. Since the objective is to compare the two FV-WENO schemes, the following test-cases have been chosen as the same as in [16].

In the present work, viscous effects are not considered in the fluid flow, and free-slip solid boundary condition is adopted by means of mirroring the reconstructed state on the solid wall, as done in [16]. Notice that for the cells near the solid wall, less stencils are used in the WENO reconstruction step.

### 6.1. Advection of a multi-phase vortex

The objective of the first test-case is to verify that the proposed scheme is formally 2nd-order accurate in space. To this end, we have chosen the test-case of the advection of a multi-phase vortex proposed in [16], of which the initial configuration in the moving frame is shown in Fig. 3 and imposed as

$$\begin{cases}\alpha_1(r,\theta,0) = (1-2\varepsilon_\alpha)\exp\left(-\dfrac{r^2}{r_0^2}\right) + \varepsilon_\alpha,\\ p(r,\theta,0) = P\left(1 - \exp\left(-\dfrac{r^2}{r_0^2}\right)\right),\\ u(r,\theta,0) = U - \sin\theta\sqrt{\dfrac{r}{\rho}\dfrac{\partial p}{\partial r}},\\ v(r,\theta,0) = V + \cos\theta\sqrt{\dfrac{r}{\rho}\dfrac{\partial p}{\partial r}},\end{cases} \tag{20}$$

where $r$ and $\theta$ are the polar coordinates, $r_0 = 0.2L$ with $L = 1$ m being the half-size of the domain, $P = 0.1\rho_{2,0}c_2^2$ with $c_1 = c_2 = 30$ m/s and $U = V = 2$ m/s.

Four mesh resolutions have been used in this convergence study: 64 × 64, 128 × 128, 256 × 256 and 512 × 512. With four periodic boundaries, at $t = 1$ s, the multi-phase vortex got back to the initial location. The errors and convergence order have been computed using the cell-average density of the mixture, which are presented in Table 2. Additionally, Fig. 4 shows the convergence behaviors of the present scheme 'Prim-1G' and the scheme 'Cons-2G' proposed in [16].

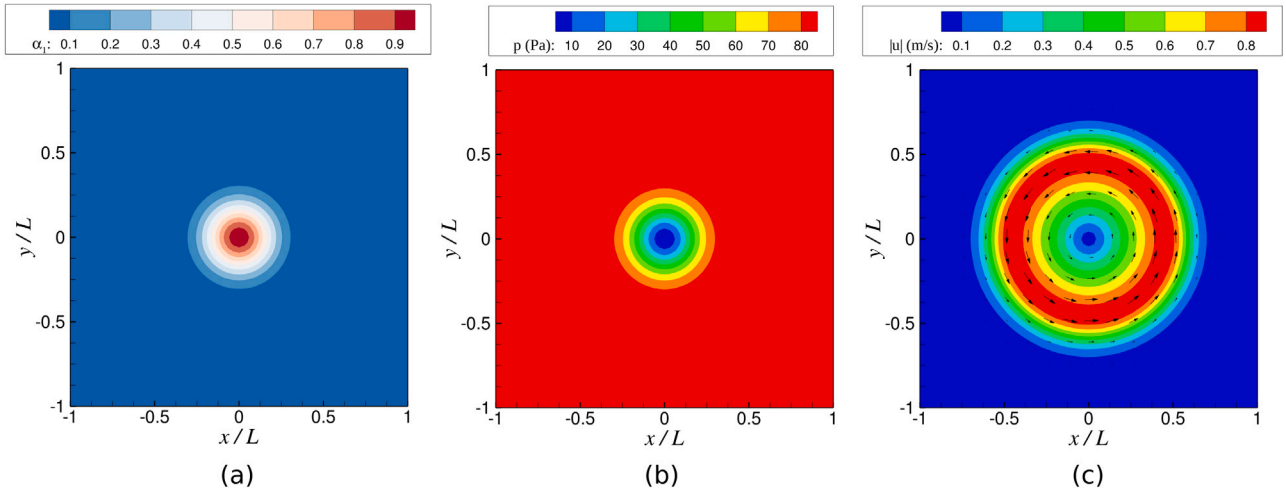


**Fig. 3.** Initial configuration of the multi-phase vortex advection test-case: (a) volume fraction field; (b) pressure field; (c) velocity field.

**Table 2**
Numerical errors in the 2D multiphase vortex advection test-case.

| Number of cells | | $L_1$-Error | $L_1$-Order | $L_2$-Error | $L_2$-Order | $L_\infty$-Error | $L_\infty$-Order |
|---|---|---|---|---|---|---|---|
| 64 × 64 | Cons-2G | 3.9112e−4 | – | 1.2847e−3 | – | 8.8276e−3 | – |
| | Prim-1G | 2.4700e−3 | – | 7.5090e−3 | – | 4.7878e−2 | – |
| 128 × 128 | Cons-2G | 1.9138e−5 | 4.3531 | 6.0041e−5 | 4.4193 | 4.5100e−4 | 4.2908 |
| | Prim-1G | 6.3501e−4 | 1.9597 | 1.9196e−3 | 1.9678 | 1.2402e−2 | 1.9488 |
| 256 × 256 | Cons-2G | 6.2019e−7 | 4.9476 | 1.9616e−6 | 4.9358 | 1.5642e−5 | 4.8496 |
| | Prim-1G | 1.5969e−4 | 1.9915 | 4.7983e−4 | 2.0002 | 4.6774e−3 | 1.4068 |
| 512 × 512 | Cons-2G | 1.7145e−8 | 5.1769 | 5.6494e−8 | 5.1178 | 4.6175e−7 | 5.0822 |
| | Prim-1G | 3.9455e−5 | 2.0170 | 1.1932e−4 | 2.0077 | 1.5923e−3 | 1.5546 |

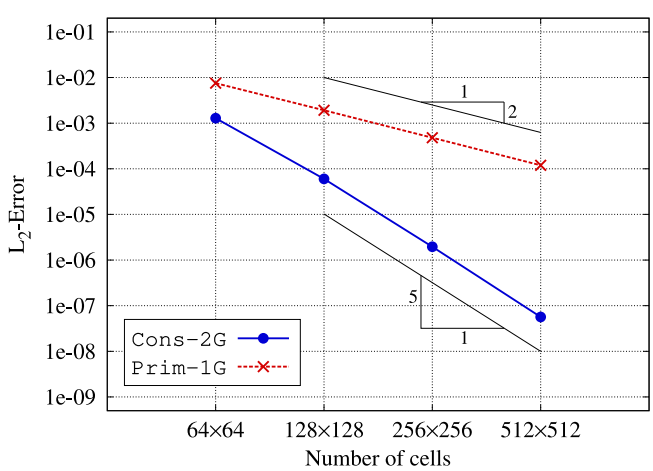


**Fig. 4.** Convergence order study in the multi-phase vortex advection test-case.

**Table 3**
Comparison between the linearized and HLLC-type Riemann flux solvers for the 'Cons-2G' scheme in the 2D multiphase vortex advection test-case.

| Number of cells | | $L_1$-Error | $L_2$-Error | $L_\infty$-Error |
|---|---|---|---|---|
| 64 × 64 | Linearized | 3.9112e−4 | 1.2847e−3 | 8.8276e−3 |
| | HLLC-type | 3.9115e−4 | 1.2848e−3 | 8.8535e−3 |
| 128 × 128 | Linearized | 1.9138e−5 | 6.0041e−5 | 4.5100e−4 |
| | HLLC-type | 1.9139e−5 | 6.0041e−5 | 4.5100e−4 |
| 256 × 256 | Linearized | 6.2019e−7 | 1.9616e−6 | 1.5642e−5 |
| | HLLC-type | 6.2019e−7 | 1.9616e−6 | 1.5642e−5 |
| 512 × 512 | Linearized | 1.7145e−8 | 5.6494e−8 | 4.6175e−7 |
| | HLLC-type | 1.7145e−8 | 5.6494e−8 | 4.6175e−7 |

In the comparisons given in Table 2 and Fig. 4, it is clearly shown that, as expected, the present scheme 'Prim-1G' is 2nd-order accurate, whereas the scheme 'Cons-2G' is 5th-order accurate. The lower order of convergence is due to the two choices made in the present work: (1) reconstruction of the primitive variables; (2) use the WENO reconstruction only once per direction.

Notice that since different approximate Riemann flux solvers have been used (HLLC-type for 'Cons-2G' and linearized one for 'Prim-1G'), it is necessary to verify that it was not the HLLC-type Riemann flux solver that degraded the convergence order. To this end, we replaced the linearized Riemann flux solver by the present HLLC-type one in the 'Cons-2G' scheme and carried out once again the convergence study. As shown in Table 3, within the 'Cons-2G' scheme, the two approximate Riemann flux solvers gave almost the same errors. Here, it is important to note that, as summarized at the end of Section 3, although the proposed scheme 'Prim-1G' sacrifices the convergence order, it is not a conventional 2nd-order accurate scheme, because it applies the 5th-order WENO reconstruction and in some special cases, e.g. where the mixture's density is almost constant or when $\mathbf{A} = \partial\mathbf{F}/\partial\mathbf{W}$ is almost constant, it can still provide a higher-order accuracy. Moreover, in the aimed applications, i.e. weakly-compressible free-surface flows, it is often expected to maintain the phase-interface as sharp as possible (unlike in this convergence study). It means that, in most of the time, only the volume fraction field $\alpha_1$ has a steep gradient across the interface. Hence, it is better to reconstruct the primitive variables, as more stencils are activated for the WENO scheme near the interface for smoother variables such as $\rho_1$, $u$ and $v$.

In order to check the interface smearing, we have carried out a similar test-case with a sharp initial $\alpha_1$ field. The results and analysis can be found in Appendix B.

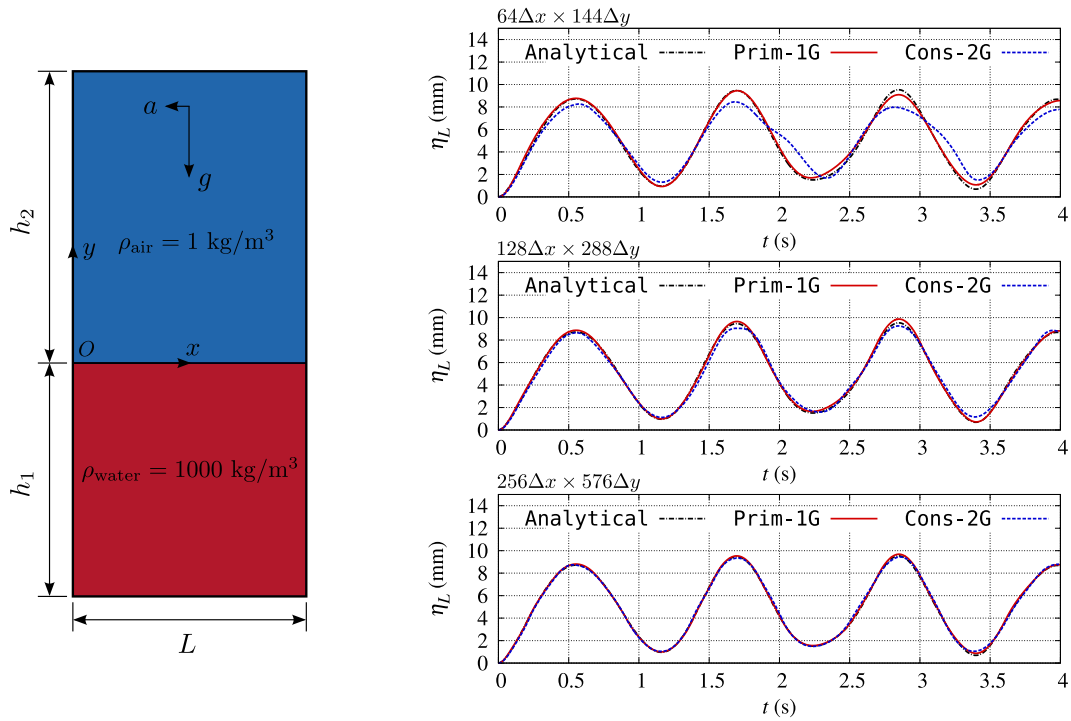


**Fig. 5.** Linear sloshing in tank test-case: the initial configuration (left) and the mesh convergence results (right) about the free-surface elevation $\eta_L = \eta(L, t)$.

**Table 4**
CPU-times for the 'Cons-2G' and 'Prim-1G' schemes in the linear sloshing test-case, which are measured at $t = 4$ s with the mesh resolution $256\Delta x \times 576\Delta y$.

| Schemes | Number of cores | CPU-time (s) | CPU-time per core (s) |
|---|---|---|---|
| Cons-2G | 24 | 5.6309e4 | 1.3514e6 |
| Prim-1G | 24 | 2.5666e4 | 6.1598e5 |

### 6.2. Linear sloshing in a rectangular tank

Next, we consider a sloshing flow inside a rectangular tank of width $L = 1$ m with free-slip walls, as shown in Fig. 5. Initially, both air and water are at rest under the effect of gravity $g = 9.81$ m/s$^2$, and $h_1 = 1$ m and $h_2 = 1.25$ m denote the initial height of the air and water domains, respectively. In addition, the artificial speeds of sound are chosen as $c_1 = c_2 = 100$ m/s. As long as $t > 0$, a horizontal acceleration $a = 0.01g$ is applied on the whole computational domain. As the amplitude is small enough, the sloshing wave can be considered as linear, of which the analytical solution is given in [22,30].

In the present study, we have carried out a series of simulations for $t = 4$ s with three mesh resolutions: $64\Delta x \times 144\Delta y$, $128\Delta x \times 288\Delta y$ and $256\Delta x \times 576\Delta y$. The evolutions in time of the free-surface elevation $\eta_L = \eta(L, t)$ at $x = L$ are shown in Fig. 5. In this comparison, one may observe that 'Prim-1G' provides a better result than 'Cons-2G' with the coarsest mesh ($64\Delta x \times 144\Delta y$), although they both converge to the analytical solution as the mesh is refined. As for the computational cost, in this specific test-case, as shown in Table 4, 'Prim-1G' is approximately twice as fast as 'Cons-2G'. Based on the above observations, it can then be concluded that, in this specific type of applications, 'Prim-1G' is not only computationally less demanding but also more accurate than 'Cons-2G' with coarser meshes. Although it sounds surprising that a higher-order scheme is less accurate than a lower-order one in this specific test-case, this can be explained by the fact that only $\alpha_1$ has a steep gradient across the air–water interface in this linear sloshing test-case, hence more stencils have been activated for smoother fields in the WENO reconstruction of 'Prim-1G' near the free-surface. In addition, the multiple sweeps in the WENO reconstruction of 'Cons-2G' may aggravate this effect.

### 6.3. Propagation of non-linear free-surface waves

To further assess the proposed numerical scheme 'Prim-1G' in the case of non-linear free-surface waves, we choose a long-time propagation of free-surface waves as the next test-case [16,31]. The initial solution is imposed by means of the stream function wave theory [32, 33]. As shown in Fig. 6, we consider a channel flow with periodic inlet/outlet boundaries (left and right) and free-slip upper and lower walls. Initially, the wave height is $H = 0.0576$ m and the wave length is $\lambda = 0.8082$ m. The length and height of the computational domain are $10\lambda$ and $2h$, respectively, where $h = 0.6$ m denotes the water depth. In addition, the artificial speeds of sound are chosen as $c_1 = c_2 = 30$ m/s.

In the present work, the computational domain is discretized with a uniform mesh of size $512\Delta x \times 512\Delta y$, which leads to $\lambda = 51.2\Delta x$ and $H = 24.576\Delta y$ with $\Delta x/\Delta y = 6.735$. The total physical time is set as $40T$ with $T = 0.7018$ s being the wave period.

As time evolves, the free-surface elevation in the middle of the domain $\eta_m(t) = \eta(x_m, t)$ with $x_m = 5\lambda$ is recorded and shown in Fig. 7, where the analytical solution is obtained with the stream function wave theory [32,33], 'WENO-Cons-2G' denotes the result in [16], and 'MUSCL-Prim-1G' and 'WENO-Prim-1G' are respectively the results obtained with the MUSCL and WENO schemes reconstructing the primitive variables only once per direction. In the present work, 'MUSCL-Prim-1G' and 'WENO-Prim-1G' are obtained with a coarser mesh than 'WENO-Cons-2G' ($1024\Delta x \times 512\Delta y$) in $x$-direction.

In addition, the fast Fourier transform (FFT) with a moving window of size $T$ has been carried out on the free-surface elevation, of which the first and second harmonic amplitudes are compared in Fig. 8.

As shown by the temporal signals given in Fig. 7 and the FFT results in Fig. 8, one may observe that: (i) 'WENO-Prim-1G' gives the closest solution to the analytical one; (ii) although both 'WENO-Cons-2G' and 'WENO-Prim-1G' introduce little numerical diffusion, 'WENO-Cons-2G' has a larger phase-error than 'WENO-Prim-1G'; (iii) 'MUSCL-Prim-1G' gives the worst result and introduces the most numerical diffusion; (iv)

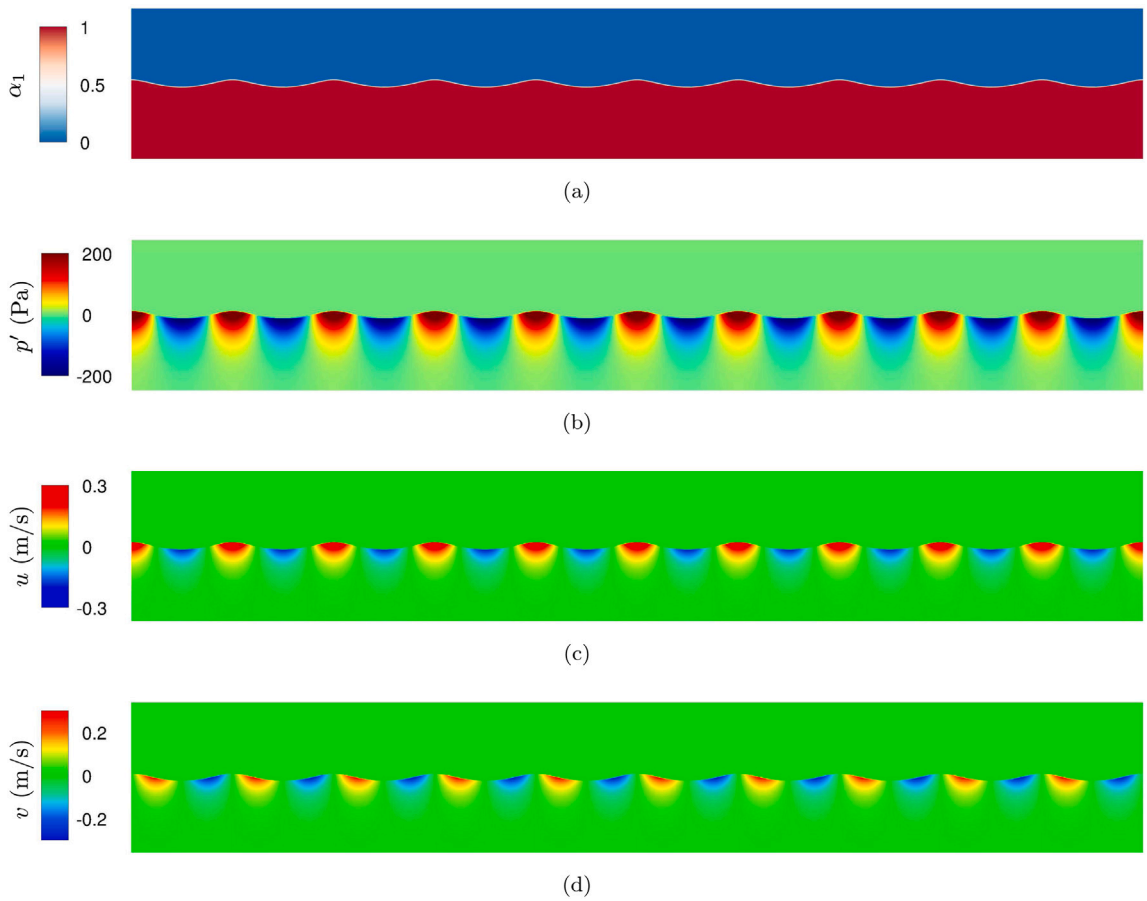


**Fig. 6.** Initial solution of the wave propagation test-case: (a) volume fraction of water $\alpha_1$; (b) relative pressure in water $p' = p + \rho_{1,0} g y$; (c) horizontal velocity component $u$; (d) vertical velocity component $v$.

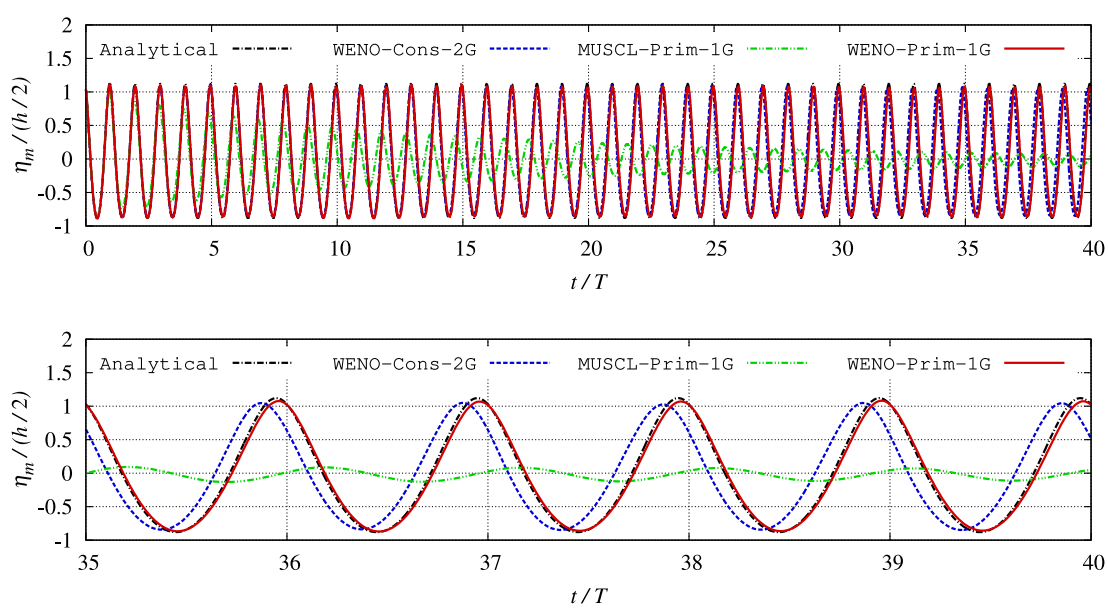


**Fig. 7.** Propagation of non-linear free-surface waves test-case: time-histories of the free-surface elevation in the middle of the domain.

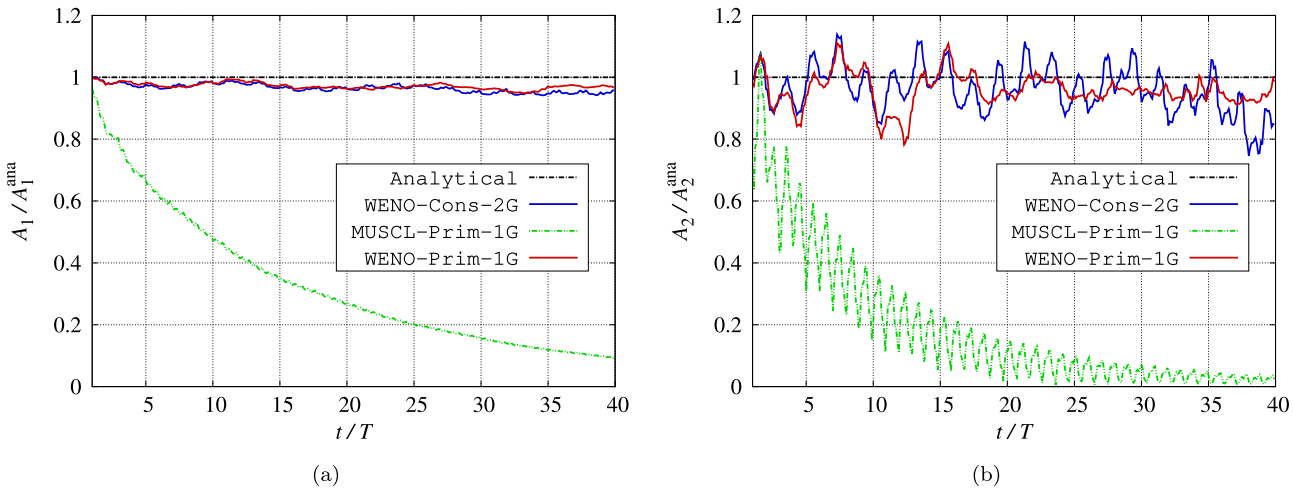


**Fig. 8.** Moving-window fast Fourier transform of the free-surface elevation $\eta_m(t) = \eta(L/2, t)$: (a) 1st-harmonic amplitude $A_1/A_1^{\rm ana}$ with $A_1^{\rm ana} \simeq 2.8166 \times 10^{-2}$ m; (b) 2nd-harmonic amplitude $A_2/A_2^{\rm ana}$ with $A_2^{\rm ana} \simeq 3.3283 \times 10^{-3}$ m.

although 'WENO-Prim-1G' is formally 2nd-order accurate, it appears to be superior to a conventional 2nd-order accurate scheme such as 'MUSCL-Prim-1G' adopted in the present test-case.

It is here worth noting that the artificial sound speed for each phase is set as 30 m/s, which is sufficiently high for satisfying the weakly-compressible hypothesis. As indicated in [16], when $\alpha_1 \simeq 0.5$, the sound speed in the two-phase mixture is much less than that of each phase, which may lead to locally 'supersonic' flows. Hence, decreasing the artificial sound speed is a way for assessing the robustness of the numerical scheme. Using the present test-case, we have tested the robustness of the linearized and HLLC-type Riemann flux solvers, by means of decreasing the artificial sound speed until divergence of the simulation. Finally, we observed that the present HLLC-type Riemann flux solver can work with a smaller sound speed ($\sim$ 3 m/s), comparing with the linearized one ($\sim$ 5 m/s). This is probably due to the fact that the HLLC-type Riemann flux solver is based on the Rankine–Hugoniot jump condition across waves, whereas the linearized one is based on the linearization of the generalized Riemann invariants by assuming that the density and sound speed of the mixture remain almost constant across waves.

### 6.4. Dam-break flow

At last, we apply the proposed scheme 'WENO-Prim-1G' to simulate the dam-break flow which has been studied experimentally in [34]. As shown in Fig. 9, a water column of height 0.3 m and width 0.6 m is initially at rest and located at the left-lower corner of a free-slip-wall container. The whole computational domain has a size of 1.61 m and 0.805 m in $x$- and $y$-directions, respectively, and is discretized with a mesh of size $512\Delta x \times 256\Delta y$. At the right-lower corner, there are four probes ($P_1 - P_4$) for recording the pressure values as time evolves. In addition, the artificial sound speeds are chosen as $c_1 = c_2 = 100$ m/s.

As shown by the flow pattern in Fig. 10, due to the gravity effect, the water column collapses and impacts on the vertical wall on the right-hand side. It can be observed that the simulated physical problem is more complex, involving more significant deformations of free-surface than the previous test-cases. Moreover, one may observe that the phase-interface represented by the volume fraction field remains generally quite thin during the simulation. Additionally, the velocity field near the instant when the topological change of the flow pattern occurs are shown in Fig. 11. It can be observed that the velocity field has locally great values at the impact moment, when the topology of the free-surface changes.

In order to further validate the proposed 'WENO-Prim-1G' FVM scheme, we compare the time evolution of the four pressure signals with the experimental measurements in [34] and the numerical results obtained with two other different types of CFD methods: the single-phase free-surface LBM [35] and the SPH method with diffusive terms [36]. As shown in the comparison given in Fig. 12, an excellent agreement can be observed among the numerical results obtained with SPH, LBM and FVM, even though they are significantly heterogeneous CFD methods. The present results agree globally well with the experimental measurements, especially for the peak values and instants recorded by the sensors $P_1$ and $P_2$ as shown in (a) and (b) of Fig. 12.

It is interesting to point out that much less numerical oscillation has been encountered in the present work with 'WENO-Prim-1G', compared with the 'WENO-Cons-2G' in the previous work [16], as shown in Fig. 13. The major reason is that, by reconstructing the primitive variables instead of the conservative ones, only the volume fraction field possesses steep gradients. As a consequence, the protection scheme [23] limiting the variation of variables (with respect to the cell-average values) was not required in the present work with 'WENO-Prim-1G'. The only adopted limiter verifies the validity of the reconstructed volume fraction value $\alpha_1$, i.e. if the reconstructed value of $\alpha_1$ is lower than zero or greater than one, its value will be replaced by the cell-average value.

Finally, we have carried out a numerical investigation on influence of the artificial compressibility effect due to the use of artificial sound speeds. Fig. 14 shows the numerical results with different sound speeds in the two-phase dam-break flow test-case at the instant when the water front impacts the right vertical wall. At this instant near the right lower corner, the fluid velocity attains its maximum value. It can be observed that, the flow pattern presented with the volume fraction $\alpha_1$, the velocity magnitude and the pressure field are extremely close for different sound speeds. With increasing artificial phasic sound speed $c_1$ and $c_2$, the local maximum Mach number in the mixture decreases as expected. Although for $c_1 = c_2 = 100$ m/s the maximum Mach number in the mixture has a value around 0.7, it is obvious that this artificial compressibility effect is negligible, as no significant difference is present in the comparison in Fig. 14.

## 7. Conclusions

In the present work, a weakly-compressible finite volume (FV) scheme is proposed for numerical simulation of nearly-incompressible immiscible two-phase flows. The WENO reconstruction scheme is applied on fixed Cartesian FV-meshes. In the present FV-WENO scheme,

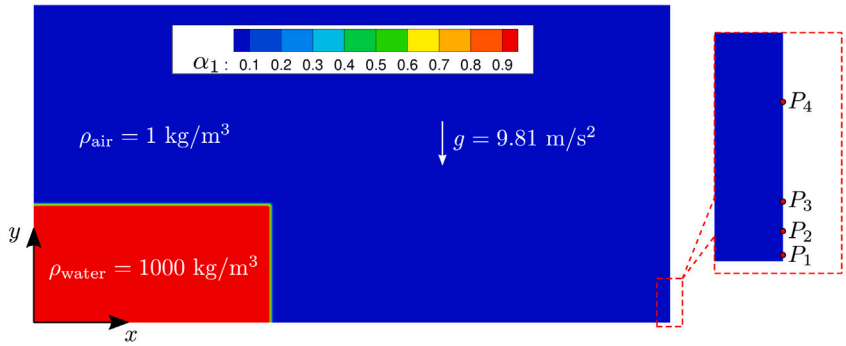


**Fig. 9.** Initial configuration of the dam-break test-case with four pressure probes: $P_1$ at $y = 3$ mm, $P_2$ at $y = 15$ mm, $P_3$ at $y = 30$ mm and $P_3$ at $y = 80$ mm.

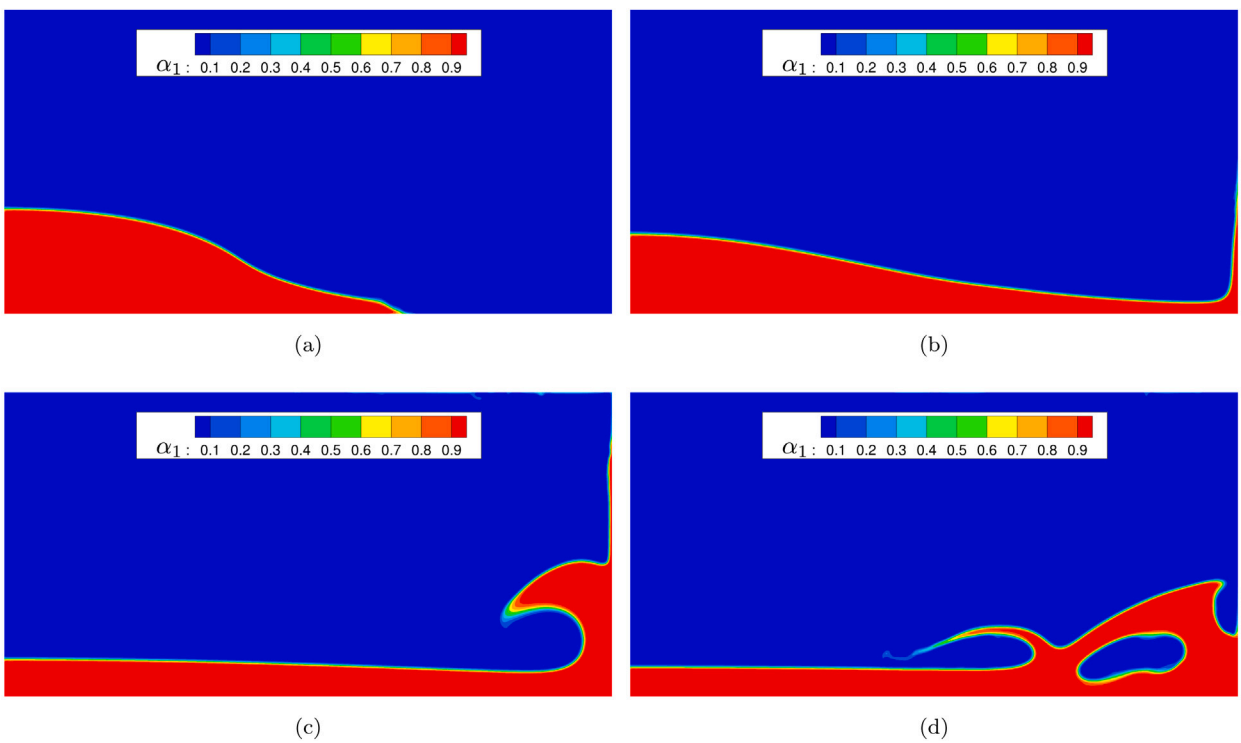


**Fig. 10.** Dam-break flow test-case: volume fraction field at four instants: (a) $t/(H/g)^{0.5} = 1.37$; (b) $t/(H/g)^{0.5} = 2.86$; (c) $t/(H/g)^{0.5} = 5.72$; (d) $t/(H/g)^{0.5} = 6.86$.

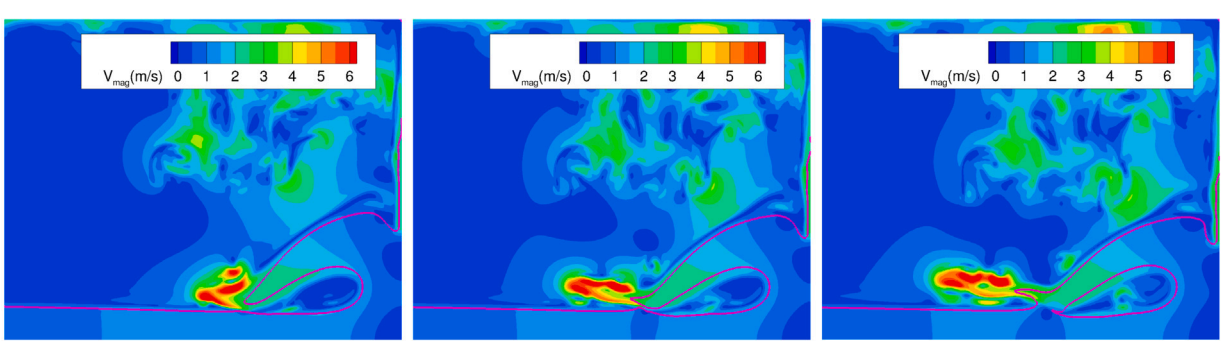


**Fig. 11.** Velocity field at three instants when the topological change occurs in the dam-break flow test-case (left: $t\sqrt{g/H} = 6.18$; middle: $t\sqrt{g/H} = 6.29$; right: $t\sqrt{g/H} = 6.40$) , where the purple curve represents the shape of the air–water interface.

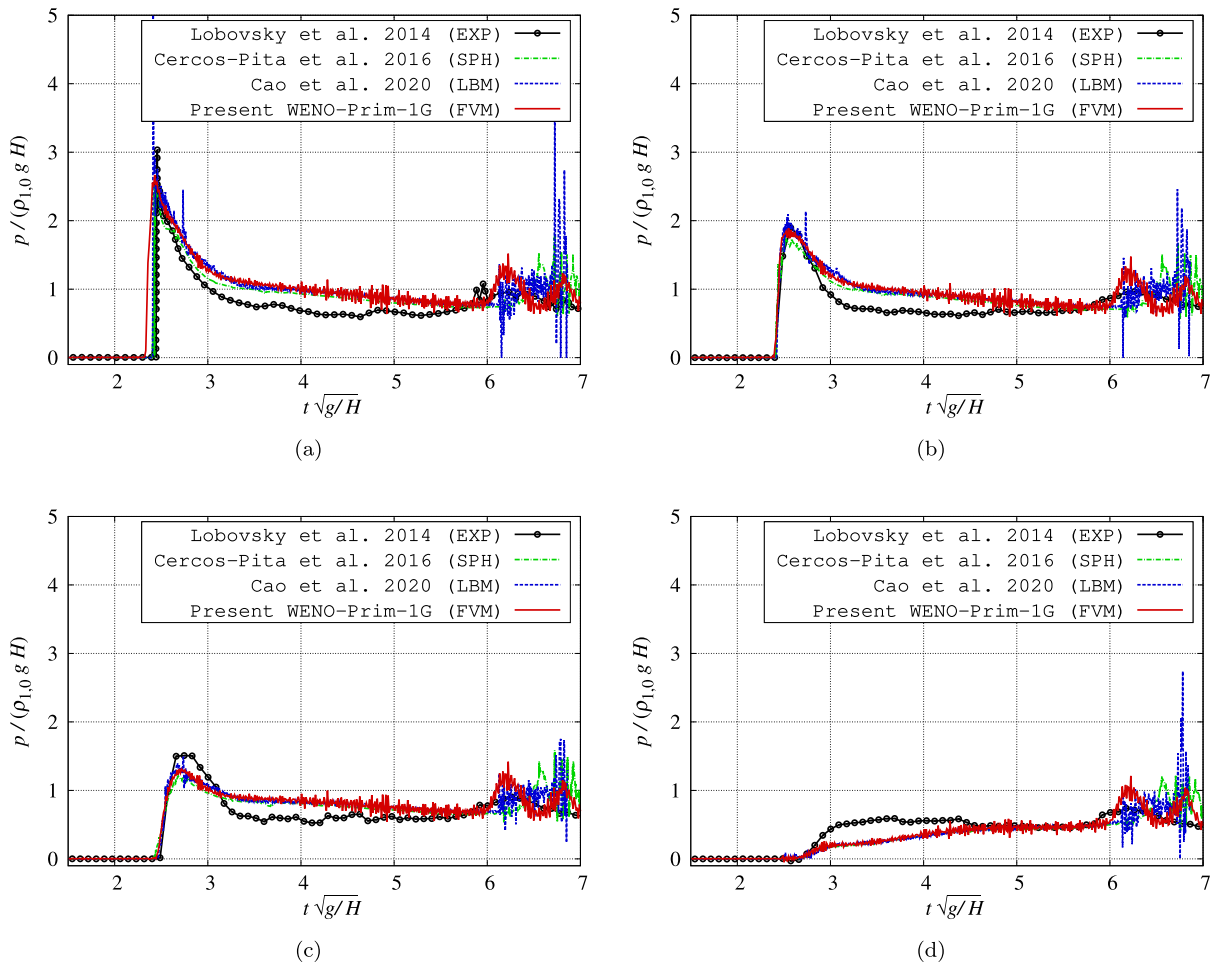


**Fig. 12.** Dam-break flow test-case: time histories of the pressure signals at the four probes as shown in Fig. 9: (a) $P_1$; (b) $P_2$; (c) $P_3$; (d) $P_4$.

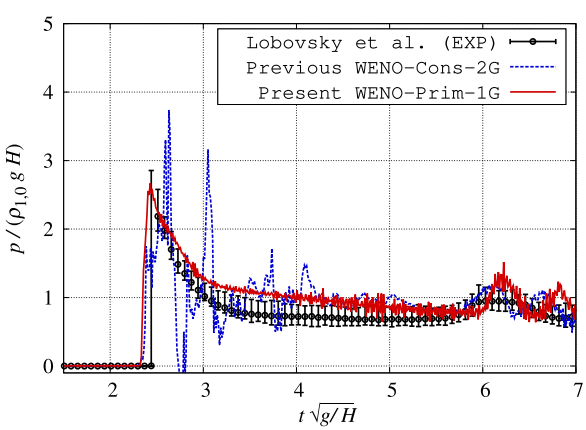


**Fig. 13.** Comparison of the recorded pressure value at the probe $P_1$ between the previous 'Cons-2G' and the present 'Prim-1G' schemes.

the computational efficiency is improved by choosing to reconstruct the primitive variables only once per direction. Reconstructing the primitive variables which are generally smoother than the conservative ones allows the use of more stencils of WENO scheme near the phase-interface. Besides, thanks to this choice, it is trivial for the proposed numerical scheme to have the oscillation-free feature in the advection of interfaces. Although the scheme is formally 2nd-order accurate in space, it is inherently different from conventional 2nd-order accurate schemes. In some cases where the mixture's density is almost constant or the governing equation is almost linear, the proposed scheme can still provide a higher-order accuracy.

Several validation test-cases have been carried out in the present work. In the advection of a multi-phase vortex test-case, it is numerically verified that the proposed scheme 'Prim-1G' is indeed 2nd-order accurate in space. Nevertheless, the comparison in the linear sloshing in tank test-case shows that, with a coarser mesh, the formally lower-order scheme 'Prim-1G' gives a more accurate evolution of the free-surface elevation than the higher-order scheme 'Cons-2G'. A similar conclusion can be obtained in the next test-case about the long-time propagation of non-linear free-surface waves. The 'surprising' observations in the linear sloshing and wave propagation test-cases are due to the fact that, in the intended applications, e.g. weakly-compressible air–water free-surface flows, only the volume fraction field possesses steep gradients, while the phase-densities and the velocity are generally smooth near the free-surface. Notice that all conservative variables possess steep gradients across the free-surface. As a consequence, more stencils for WENO reconstruction have been activated in 'Prim-1G' than in 'Cons-2G'. Finally, in the dam-break flow test-case, the recorded pressure signals are in an excellent agreement with heterogeneous CFD methods and the experimental measurements. With the reconstruction of smoother primitive variables, less spurious oscillations have been observed with the 'Prim-1G' scheme than with the 'Cons-2G' one.

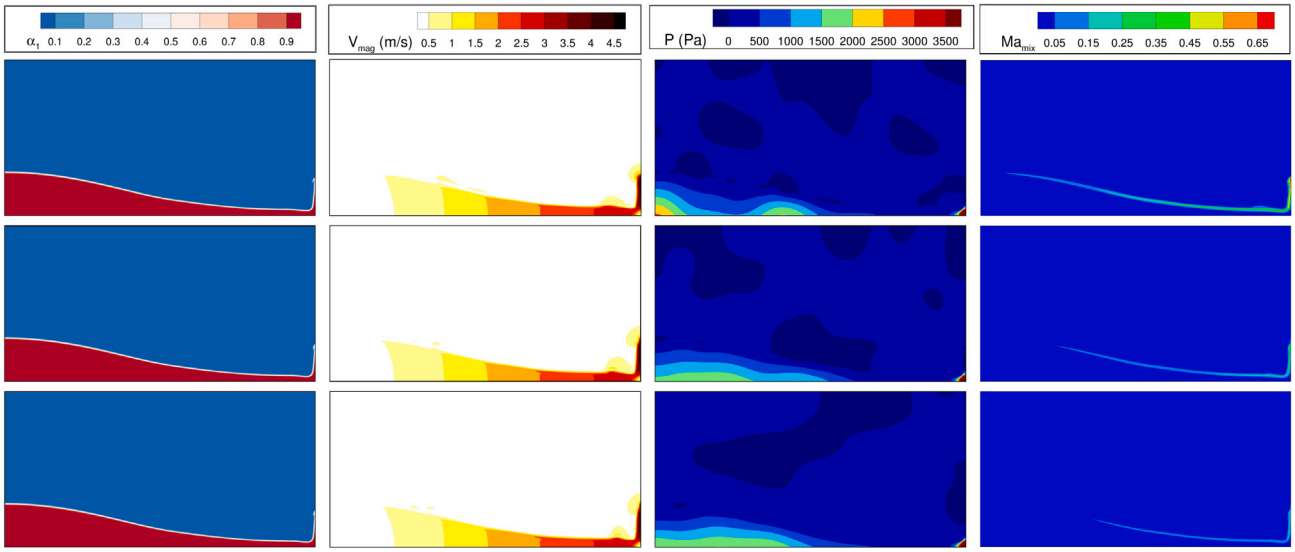


**Fig. 14.** Influence of the artificial sound speed in the two-phase dam-break flow (top-row: $c_1 = c_2 = 100$ m/s; middle-row: $c_1 = c_2 = 200$ m/s; bottom-row: $c_1 = c_2 = 300$ m/s).

## CRediT authorship contribution statement

**Zhe Li:** Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing. **Louis Vittoz:** Software, Methodology, Validation. **Guillaume Oger:** Software, Methodology, Formal analysis, Writing – review & editing. **David Le Touzé:** Methodology, Formal analysis, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix A. A straightforward way to obtain the cell-average primitive variables

In the following, we shall show that the operations in Eqs. (8) and (9) induce generally an error of order $\mathcal{O}(\Delta x^2)$. To do so, we consider a 1D example and give the following proposition.

**Proposition 1.** *Given the numerical cell-average values* $\overline{(\xi)}_i^{\text{num}} = \overline{(\xi)}_i^{\text{ana}} + \mathcal{O}(\Delta x^m)$ *and* $\overline{(\xi\varphi)}_i^{\text{num}} = \overline{(\xi\varphi)}_i^{\text{ana}} + \mathcal{O}(\Delta x^m)$ *with* $m \geq 2$, *the cell-average value of* $\varphi$ *for the* $i$*th cell can be approximated as*

$$\widetilde{(\varphi)}_i = \frac{\overline{(\xi\varphi)}_i^{\text{num}}}{\overline{(\xi)}_i^{\text{num}}} = \overline{(\varphi)}_i^{\text{ana}} + \mathcal{O}(\Delta x^2), \tag{A.1}$$

*which means the error of* $\widetilde{(\varphi)}_i$, *compared to its analytical cell-average value* $\overline{(\varphi)}_i^{\text{ana}}$, *is of order* $\mathcal{O}(\Delta x^2)$.

**Proof.** Using the definition given in Eq. (A.1), one has

$$\widetilde{(\varphi)}_i = \frac{\overline{(\xi\varphi)}_i^{\text{num}}}{\overline{(\xi)}_i^{\text{num}}} = \frac{\overline{(\xi\varphi)}_i^{\text{ana}} + \mathcal{O}(\Delta x^m)}{\overline{(\xi)}_i^{\text{ana}} + \mathcal{O}(\Delta x^m)} = \frac{\overline{(\xi\varphi)}_i^{\text{ana}}}{\overline{(\xi)}_i^{\text{ana}}} + \mathcal{O}(\Delta x^m). \tag{A.2}$$

With the Taylor expansion of $\xi(x)$ around $x_i$ in the definition of the cell-average value, one has

$$\begin{aligned}
\overline{(\xi)}_i^{\text{ana}} &= \frac{1}{\Delta x}\int_{x_{i-1/2}}^{x_{i+1/2}} \xi_i + \frac{(x-x_i)}{1!}\xi_i^{(1)} + \frac{(x-x_i)^2}{2!}\xi_i^{(2)} + \cdots \, \mathrm{d}x, \\
&= \frac{\xi_i}{\Delta x}\int_{x_{i-1/2}}^{x_{i+1/2}} \mathrm{d}x + \frac{\xi_i^{(1)}}{\Delta x}\int_{x_{i-1/2}}^{x_{i+1/2}} \frac{(x-x_i)}{1!}\,\mathrm{d}x \\
&\quad + \frac{\xi_i^{(2)}}{\Delta x}\int_{x_{i-1/2}}^{x_{i+1/2}} \frac{(x-x_i)^2}{2!}\,\mathrm{d}x + \ldots, \\
&= \xi_i + \frac{\xi_i^{(2)}}{\Delta x}\left(\frac{(x_{i+1/2}-x_i)^3}{6} - \frac{(x_{i-1/2}-x_i)^3}{6}\right) + \cdots, \\
&= \xi_i + \mathcal{O}(\Delta x^2),
\end{aligned} \tag{A.3}$$

where $\xi_i$ and $\xi_i^{(n)}$ denote respectively the analytical value and the $n$th-order derivative of $\xi(x)$ at $x_i$ and $x_{i\pm1/2} = x_i \pm \Delta x/2$. Similarly, one can have $\overline{(\xi\varphi)}_i^{\text{ana}} = \xi_i\varphi_i + \mathcal{O}(\Delta x^2)$.

Now, substituting $\overline{(\xi)}_i^{\text{ana}} = \xi_i + \mathcal{O}(\Delta x^2)$ and $\overline{(\xi\varphi)}_i^{\text{ana}} = \xi_i\varphi_i + \mathcal{O}(\Delta x^2)$ into Eq. (A.2) yields

$$\widetilde{(\varphi)}_i = \frac{\overline{(\xi\varphi)}_i^{\text{ana}}}{\overline{(\xi)}_i^{\text{ana}}} + \mathcal{O}(\Delta x^m) = \frac{\xi_i\varphi_i + \mathcal{O}(\Delta x^2)}{\xi_i + \mathcal{O}(\Delta x^2)} + \mathcal{O}(\Delta x^m) = \varphi_i + \mathcal{O}(\Delta x^2) = \overline{(\varphi)}_i^{\text{ana}} + \mathcal{O}(\Delta x^2), \tag{A.4}$$

where $m \geq 2$ and we have applied $\overline{(\varphi)}_i^{\text{ana}} = \varphi_i + \mathcal{O}(\Delta x^2)$. □

## Appendix B. A pure advection test-case

In order to check the interface smearing, we have carried out a pure advection test-case of disk with a sharp initial $\alpha_1$ field, which was recently conducted by De Vuyst et al. [37].

The configuration of this test-case is quite similar to the one presented in Section 6.1, except that the initial pressure is set as zero everywhere and a sharp initial condition is imposed for the $\alpha_1$ field. In addition, the initial velocity field is homogeneous with $u = 1$ m/s and $v = 1$ m/s. The computational domain is discretized with a uniform mesh of size $256 \times 256$. The artificial sound speeds are set as $c_1 = c_2 = 25$ m/s.

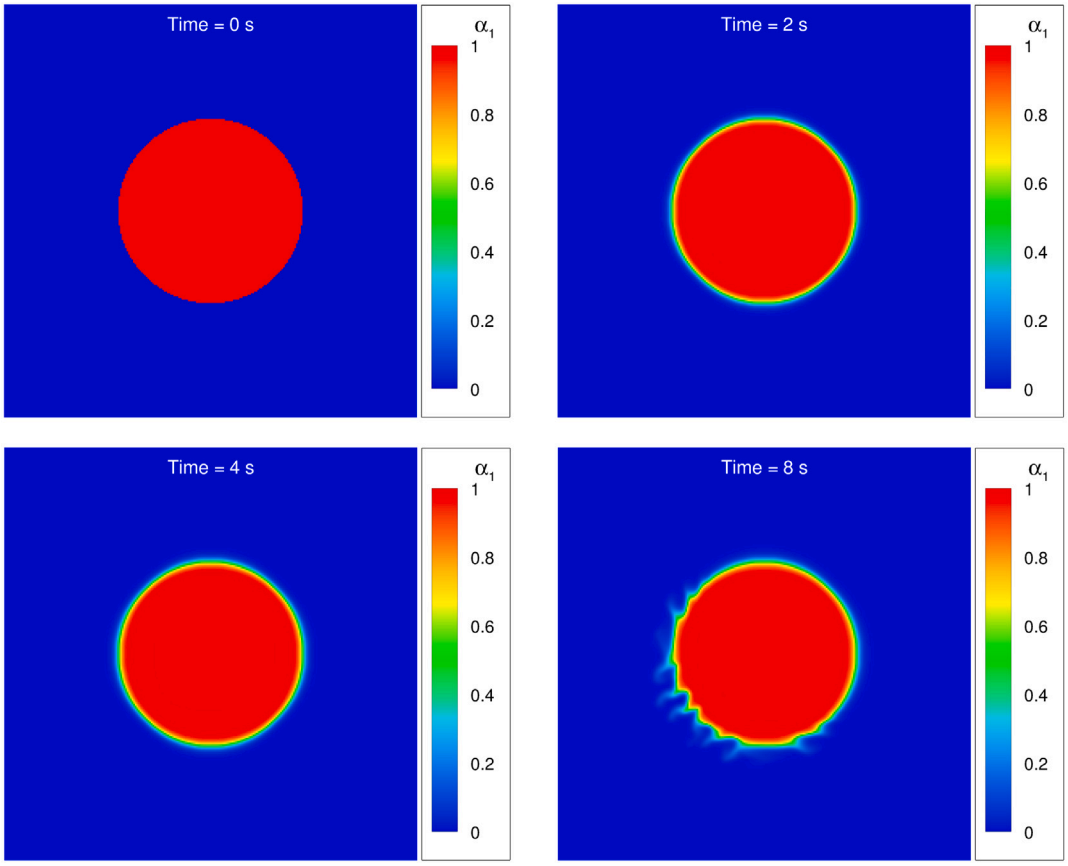


**Fig. B.1.** $\alpha_1$ field at four instants in the pure advection test-case.

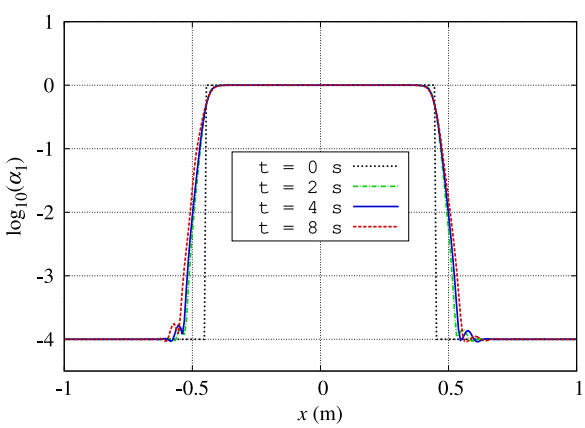


**Fig. B.2.** The profile of $\log_{10}(\alpha_1)$ at the middle cross-section in the pure advection test-case.

Fig. B.1 shows the $\alpha_1$ field at four instants: $t = 0, 2, 4$ and $8$ s, corresponding to the initial instant, 1, 2 and 4 periods of advection, respectively. The profile of $\log_{10}(\alpha_1)$ at the middle cross-section is provided in Fig. B.2. From these results one can observe that, as time evolves, the phase-interface has been smeared slightly, the shape of disk has been well maintained and the diffuse interface remained quite thin until two periods of advection. However, at the end of the fourth period, the numerical smearing became significant and the shape of the disk contained obvious noises.

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